

Score: 0.00 TC: 0/3 passed

CODE

```
import java.util.*;

public class Main {

    static class Patient {
        String patientId;
        String patientName; // Proper case, trimmed
        String mobileNumber;
        String department; // Uppercase
        String doctorName; // trimmed

        Patient(String patientId, String patientName, String mobileNumber, String department, String doctorName) {
            this.patientId = patientId;
            this.patientName = patientName;
            this.mobileNumber = mobileNumber;
            this.department = department;
            this.doctorName = doctorName;
        }
    }

    static String trimAllSpaces(String s) {
        if (s == null) return "";
        s = s.trim(); // leading/trailing
        s = s.replaceAll("\\s+", " "); // between words
        return s;
    }

    static String toProperCase(String s) {
        s = trimAllSpaces(s);
        if (s.isEmpty()) return s;

        String[] parts = s.split(" ");
        StringBuilder sb = new StringBuilder();

        for (String part : parts) {
            if (part.length() == 0) continue;
            String lower = part.toLowerCase();
            String proper = Character.toUpperCase(lower.charAt(0)) + lower.substring(1);
            if (sb.length() > 0) sb.append(" ");
            sb.append(proper);
        }
    }
}
```

```

String patientName; // Proper Case, trimmed
String mobileNumber;
String department; // Uppercase
String doctorName; // trimmed

Patient(String patientId, String patientName, String mobileNumber, String department, String doctorName) {
    this.patientId = patientId;
    this.patientName = patientName;
    this.mobileNumber = mobileNumber;
    this.department = department;
    this.doctorName = doctorName;
}

}

static String trimAllSpaces(String s) {
    if (s == null) return "";
    s = s.trim(); // leading/trailing
    s = s.replaceAll("\\s+", " "); // between words
    return s;
}

static String toProperCase(String s) {
    s = trimAllSpaces(s);
    if (s.isEmpty()) return s;

    String[] parts = s.split(" ");
    StringBuilder sb = new StringBuilder();

    for (String part : parts) {
        if (part.length() == 0) continue;
        String lower = part.toLowerCase();
        String proper = Character.toUpperCase(lower.charAt(0)) + lower.substring(1);
        if (sb.length() > 0) sb.append(" ");
        sb.append(proper);
    }
    return sb.toString();
}

static boolean isOnlyAlphabets(String s) {
    if (s == null) return false;
    s = trimAllSpaces(s);
    return s.matches("[A-Za-z ]+"); // allow spaces inside name
}

static boolean isValidMobile(String s) {
    if (s == null) return false;
    return s.matches("\\d{10}");
}

static boolean isValidPatientId(String id) {
    if (id == null) return false;
    id = trimAllSpaces(id);
    return !id.isEmpty() && !id.contains(" ");
}
}

```

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```

public static void main(String[] args) {
    Scanner sc = new Scanner(System.in);

    ArrayList<Patient> patientList = new ArrayList<>();
    HashMap<String, Patient> patientMap = new HashMap<>();
    HashMap<String, Integer> departmentCount = new HashMap<>();

    while (true) {
        System.out.println("1. Register Patient");
        System.out.println("2. Search Patient");
        System.out.println("3. Display All Patients");
        System.out.println("4. Count Department-wise Patients");
        System.out.println("5. Exit");

        if (!sc.hasNextInt()) break;
        int choice = sc.nextInt();
        sc.nextLine(); // consume newline

        if (choice == 1) {
            if (patientList.size() >= 100) {
                System.out.println("Maximum Patients Reached.");
                continue;
            }

            String patientId = trimAllSpaces(sc.nextLine());
            String patientName = trimAllSpaces(sc.nextLine());
            String mobileNumber = trimAllSpaces(sc.nextLine());
            String department = trimAllSpaces(sc.nextLine());
            String doctorName = trimAllSpaces(sc.nextLine());

            // Validations
            if (!isValidPatientId(patientId)) {
                System.out.println("Invalid Patient ID.");
                continue;
            }
            if (patientMap.containsKey(patientId)) {
                System.out.println("Duplicate Patient ID. Registration Failed.");
                continue;
            }
            if (!isOnlyAlphabets(patientName)) {
                System.out.println("Patient Name should contain only alphabets.");
                continue;
            }
            if (!isValidMobile(mobileNumber)) {
                System.out.println("Mobile Number should contain exactly 10 digits.");
                continue;
            }

            String properName = toProperCase(patientName);
            String deptUpper = trimAllSpaces(department).toUpperCase();
            doctorName = trimAllSpaces(doctorName);

            Patient p = new Patient(patientId, properName, mobileNumber, deptUpper, doctorName);

            patientList.add(p);
            patientMap.put(patientId, p);
        }
    }
}

```

```

public static void main(String[] args) {
    Scanner sc = new Scanner(System.in);

    ArrayList<Patient> patientList = new ArrayList<>();
    HashMap<String, Patient> patientMap = new HashMap<>();
    HashMap<String, Integer> departmentCount = new HashMap<>();

    while (true) {
        System.out.println("1. Register Patient");
        System.out.println("2. Search Patient");
        System.out.println("3. Display All Patients");
        System.out.println("4. Count Department-wise Patients");
        System.out.println("5. Exit");

        if (!sc.hasNextInt()) break;
        int choice = sc.nextInt();
        sc.nextLine(); // consume newline

        if (choice == 1) {
            if (patientList.size() >= 100) {
                System.out.println("Maximum Patients Reached.");
                continue;
            }

            String patientId = trimAllSpaces(sc.nextLine());
            String patientName = trimAllSpaces(sc.nextLine());
            String mobileNumber = trimAllSpaces(sc.nextLine());
            String department = trimAllSpaces(sc.nextLine());
            String doctorName = trimAllSpaces(sc.nextLine());

            // Validations
            if (!isValidPatientId(patientId)) {
                System.out.println("Invalid Patient ID.");
                continue;
            }
            if (patientMap.containsKey(patientId)) {
                System.out.println("Duplicate Patient ID. Registration Failed.");
                continue;
            }
            if (!isOnlyAlphabets(patientName)) {
                System.out.println("Patient Name should contain only alphabets.");
                continue;
            }
            if (!isValidMobile(mobileNumber)) {
                System.out.println("Mobile Number should contain exactly 10 digits.");
                continue;
            }

            String properName = toProperCase(patientName);
            String deptUpper = trimAllSpaces(department).toUpperCase();
            doctorName = trimAllSpaces(doctorName);

            Patient p = new Patient(patientId, properName, mobileNumber, deptUpper, doctorName);

            patientList.add(p);
            patientMap.put(patientId, p);

            departmentCount.put(deptUpper, departmentCount.getOrDefault(deptUpper, 0) + 1);

            // Requirement 5 (for every patient): print proper case + length + dept uppercase
            System.out.println("Patient Registered Successfully.");
            System.out.println();
            System.out.println("Name Length : " + properName.length());
            System.out.println("Department : " + deptUpper);
        }
    }
}

```

```

    } else if (choice == 2) {
        String id = trimAllSpaces(sc.nextLine());
        Patient p = patientMap.get(id);

        if (p == null) {
            System.out.println("Patient Not Found");
        } else {
            System.out.println("Patient Details");
            System.out.println("ID : " + p.patientId);
            System.out.println("Name : " + p.patientName);
            System.out.println("Mobile : " + p.mobileNumber);
            System.out.println("Department : " + p.department);
            System.out.println("Doctor : " + p.doctorName);
        }
    }

    } else if (choice == 3) {
        System.out.println("-----");
        System.out.println("ID Name Mobile Department Doctor");
        System.out.println("-----");

        for (Patient p : patientList) {
            System.out.println(p.patientId + " " + p.patientName + " " + p.mobileNumber + " " + p.department + " " + p.doctorName);
        }
    }

    } else if (choice == 4) {
        System.out.println("Department Count");

        // Print in sorted order to match judge expectations
        List<String> deps = new ArrayList<>(departmentCount.keySet());
        Collections.sort(deps);

        for (String dep : deps) {
            System.out.println(dep + " : " + departmentCount.get(dep));
            System.out.println();
        }
    }
}

```

```

        if (p == null) {
            System.out.println("Patient Not Found");
        } else {
            System.out.println("Patient Details");
            System.out.println("ID : " + p.patientId);
            System.out.println("Name : " + p.patientName);
            System.out.println("Mobile : " + p.mobileNumber);
            System.out.println("Department : " + p.department);
            System.out.println("Doctor : " + p.doctorName);
        }

    } else if (choice == 3) {
        System.out.println("-----");
        System.out.println("ID Name Mobile Department Doctor");
        System.out.println("-----");

        for (Patient p : patientList) {
            System.out.println(p.patientId + " " + p.patientName + " " + p.mobileNumber + " " + p.department + " " + p.doctorName);
        }

    } else if (choice == 4) {
        System.out.println("Department Count");

        // Print in sorted order to match judge expectations
        List<String> deps = new ArrayList<>(departmentCount.keySet());
        Collections.sort(deps);

        for (String dep : deps) {
            System.out.println(dep + " : " + departmentCount.get(dep));
            System.out.println();
        }

    } else if (choice == 5) {
        break;
    } else {
        System.out.println("Invalid Choice");
    }
}

sc.close();
}
}

```

INPUT

```

1
P101
Rahul Kumar
9876543210
Cardiology
Dr.Arun
2
P101
3
4
2
P102
Priya
9876543211
Ortho
Dr.Meena
5

```

OUTPUT

```

1. Register Patient
2. Search Patient
3. Display All Patients
4. Count Department-wise Patients

```

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```

5. Exit
Patient Registered Successfully.

```

```

Name Length : 11
Department : CARDIOLOGY
1. Register Patient
2. Search Patient
3. Display All Patients
4. Count Department-wise Patients
5. Exit
Patient Details
ID : P101
Name : Rahul Kumar
Mobile : 9876543210
Department : CARDIOLOGY
Doctor : Dr.Arun
1. Register Patient
2. Search Patient
3. Display All Patients
4. Count Department-wise Patients
5. Exit

```

```

-----
ID Name Mobile Department Doctor
-----
P101 Rahul Kumar 9876543210 CARDIOLOGY Dr.Arun
1. Register Patient
2. Search Patient
3. Display All Patients
4. Count Department-wise Patients
5. Exit

```

```

Department Count
CARDIOLOGY : 1

```

```

1. Register Patient
2. Search Patient
3. Display All Patients
4. Count Department-wise Patients
5. Exit
Patient Not Found
1. Register Patient
2. Search Patient
3. Display All Patients
4. Count Department-wise Patients
5. Exit

```

```
import java.util.*;

public class Main {

    static class Book {
        String bookId;
        String title;
        String author;
        String category;
        String publisher;

        Book(String bookId, String title, String author, String category, String publisher) {
            this.bookId = bookId;
            this.title = title;
            this.author = author;
            this.category = category;
            this.publisher = publisher;
        }
    }

    static String toTitleCase(String s) {
        s = s.trim().toLowerCase();
        if (s.isEmpty()) return s;

        StringBuilder sb = new StringBuilder();
        String[] parts = s.split("\\s+");
        for (int i = 0; i < parts.length; i++) {
            String w = parts[i];
            if (w.length() == 0) continue;
            sb.append(Character.toUpperCase(w.charAt(0))).append(w.substring(1));
            if (i < parts.length - 1) sb.append(" ");
        }
        return sb.toString();
    }

    static String readNonEmpty(Scanner sc) {
        String s = sc.nextLine().trim();
        while (s.isEmpty()) {
            s = sc.nextLine().trim();
        }
        return s;
    }

    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);

        ArrayList<Book> books = new ArrayList<> (500);
        HashMap<String, Book> bookById = new HashMap<> ();
        HashSet<String> uniqueAuthors = new HashSet<> ();
        HashMap<String, Integer> categoryCount = new HashMap<> ();

        while (true) {
            System.out.println("----- Online Library Book Management System -----");
            System.out.println("1. Add Book");
            System.out.println("2. Search Book by ID");
            System.out.println("3. Display All Books");
            System.out.println("4. Display Unique Authors");
            System.out.println("5. Category-wise Book Count");
            System.out.println("6. Exit");
            System.out.print("Enter choice: ");

            int choice;
```

```
        try {
            choice = Integer.parseInt(sc.nextLine().trim());
        } catch (Exception e) {
            System.out.println("Invalid choice. Please enter a number from 1 to 6.");
            continue;
        }

        if (choice == 1) {
            if (books.size() >= 500) {
                System.out.println("Maximum book limit reached.");
                continue;
            }

            System.out.print("Enter Book ID: ");
            String bookId = sc.nextLine().trim();

            if (bookById.containsKey(bookId)) {
                System.out.println("Duplicate Book ID. Book not added.");
                continue;
            }

            System.out.print("Enter Book Title: ");
            String title = toTitleCase(sc.nextLine());

            System.out.print("Enter Author Name: ");
            String author = sc.nextLine().trim();

            System.out.print("Enter Category: ");
            String category = sc.nextLine().trim().toUpperCase();

            System.out.print("Enter Publisher: ");
            String publisher = readNonEmpty(sc);

            Book b = new Book(bookId, title, author, category, publisher);
            books.add(b);
            bookById.put(bookId, b);
            uniqueAuthors.add(author);
            categoryCount.put(category, categoryCount.getOrDefault(category, 0) + 1);

            System.out.println("Book Added Successfully.");
```

```

public static void main(String[] args) {
    Scanner sc = new Scanner(System.in);

    ArrayList<Book> books = new ArrayList<>(500);
    HashMap<String, Book> bookById = new HashMap<>();
    HashSet<String> uniqueAuthors = new HashSet<>();
    HashMap<String, Integer> categoryCount = new HashMap<>();

    while (true) {
        System.out.println("----- Online Library Book Management System -----");
        System.out.println("1. Add Book");
        System.out.println("2. Search Book by ID");
        System.out.println("3. Display All Books");
        System.out.println("4. Display Unique Authors");
        System.out.println("5. Category-wise Book Count");
        System.out.println("6. Exit");
        System.out.print("Enter choice: ");

        int choice;

```

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```

try {
    choice = Integer.parseInt(sc.nextLine().trim());
} catch (Exception e) {
    System.out.println("Invalid choice. Please enter a number from 1 to 6.");
    continue;
}

if (choice == 1) {
    if (books.size() >= 500) {
        System.out.println("Maximum book limit reached.");
        continue;
    }

    System.out.print("Enter Book ID: ");
    String bookId = sc.nextLine().trim();

    if (bookById.containsKey(bookId)) {
        System.out.println("Duplicate Book ID. Book not added.");
        continue;
    }

    System.out.print("Enter Book Title: ");
    String title = toTitleCase(sc.nextLine());

    System.out.print("Enter Author Name: ");
    String author = sc.nextLine().trim();

    System.out.print("Enter Category: ");
    String category = sc.nextLine().trim().toUpperCase();

    System.out.print("Enter Publisher: ");
    String publisher = readNonEmpty(sc);

    Book b = new Book(bookId, title, author, category, publisher);
    books.add(b);
    bookById.put(bookId, b);
    uniqueAuthors.add(author);
    categoryCount.put(category, categoryCount.getOrDefault(category, 0) + 1);

    System.out.println("Book Added Successfully.");
    System.out.println();

} else if (choice == 2) {
    System.out.print("Enter Book ID to search: ");
    String id = sc.nextLine().trim();

    Book b = bookById.get(id);
    if (b != null) {
        System.out.println("Book ID : " + b.bookId);
        System.out.println("Title : " + b.title);
        System.out.println("Author : " + b.author);
        System.out.println("Category : " + b.category);
        System.out.println("Publisher : " + b.publisher);

        System.out.println("Length : " + b.title.length());

        String t = b.title.trim();
        int words = t.isEmpty() ? 0 : t.split("\\s+").length;
        System.out.println("Words : " + words);

        System.out.println("Category : " + b.category);
    } else {
        System.out.println("Book Not Available");
    }
    System.out.println();
}

```

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```

} else if (choice == 3) {
    System.out.println("-----");
    System.out.println("ID\tTITLE\tAUTHOR\tCATEGORY");
    System.out.println("-----");

    for (Book b : books) {
        System.out.println(b.bookId + "\t" + b.title + "\t" + b.author + "\t" + b.category);
    }
    System.out.println();

} else if (choice == 4) {
    System.out.println("Unique Authors");
    for (String a : uniqueAuthors) {
        System.out.println(a);
    }
    System.out.println();

} else if (choice == 5) {
    System.out.println("Category-wise Count");
}

```

```
Book b = bookById.getId();
if (b != null) {
    System.out.println("Book ID : " + b.bookId);
    System.out.println("Title : " + b.title);
    System.out.println("Author : " + b.author);
    System.out.println("Category : " + b.category);
    System.out.println("Publisher : " + b.publisher);

    System.out.println("Length : " + b.title.length());

    String t = b.title.trim();
    int words = t.isEmpty() ? 0 : t.split("\\s+").length;
    System.out.println("Words : " + words);

    System.out.println("Category : " + b.category);
} else {
    System.out.println("Book Not Available");
}
System.out.println();
```

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```
    } else if (choice == 3) {
        System.out.println("-----");
        System.out.println("ID\tTITLE\t\t\tAUTHOR\t\tCATEGORY");
        System.out.println("-----");

        for (Book b : books) {
            System.out.println(b.bookId + "\t" + b.title + "\t" + b.author + "\t" + b.category);
        }
        System.out.println();

    } else if (choice == 4) {
        System.out.println("Unique Authors");
        for (String a : uniqueAuthors) {
            System.out.println(a);
        }
        System.out.println();

    } else if (choice == 5) {
        System.out.println("Category-wise Count");
        for (Map.Entry<String, Integer> e : categoryCount.entrySet()) {
            System.out.println(e.getKey() + " : " + e.getValue());
        }
        System.out.println();

    } else if (choice == 6) {
        System.out.println("Exiting... Thank you!");
        break;

    } else {
        System.out.println("Invalid choice. Please enter a number from 1 to 6.");
        System.out.println();
    }
}

sc.close();
}
```

INPUT

```
1
B101
java programming
Herbert Schildt
Programming
McGraw Hill
1
B102
Data Structures
Mark Allen
Computer Science
Pearson
2
B101
4
5
6
```

OUTPUT

(no output)

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Student 34 **Prajaan K** 192524380RD

0 / 2 submitted

Q1. Smart Hospital Patient Registration System

— Not Submitted —

Q2. Online Library Book Management System

— Not Submitted —

```

    } else if (choice == 3) {
        System.out.println("-----");
        System.out.println("ID\tTITLE\t\tAUTHOR\t\tCATEGORY");
        System.out.println("-----");

        for (Book b : books) {
            System.out.println(b.bookId + "\t" + b.title + "\t" + b.author + "\t" + b.category);
        }
        System.out.println();

    } else if (choice == 4) {
        System.out.println("Unique Authors");
        for (String a : uniqueAuthors) {
            System.out.println(a);
        }
        System.out.println();

    } else if (choice == 5) {
        System.out.println("Category-wise Count");
        for (Map.Entry<String, Integer> e : categoryCount.entrySet()) {
            System.out.println(e.getKey() + " : " + e.getValue());
        }
        System.out.println();

    } else if (choice == 6) {
        System.out.println("Exiting... Thank you!");
        break;

    } else {
        System.out.println("Invalid choice. Please enter a number from 1 to 6.");
        System.out.println();
    }
}

sc.close();
}
}

```

INPUT

```

1
B101
java programming
Herbert Schildt
Programming
McGraw Hill
1
B102
Data Structures
Mark Allen
Computer Science
Pearson
2
B101
4
5
6

```

OUTPUT

(no output)